

Optical counting platform of shrimp larvae using masked k -means and a side window filter

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Accurate and efficient counting of shrimp larvae is crucial for monitoring reproduction patterns, assessing growth rates, and evaluating the performance of aquaculture. Traditional methods via density estimation are ineffective in the case of high density. In addition, the image contains bright spots utilizing the point light source or the line light source. Therefore, in this paper an automated shrimp counting platform based on optics and image processing is designed to complete the task of counting shrimp larvae. First, an area light source ensures a uniformly illuminated environment, which helps to obtain shrimp images with high resolution. Then, a counting algorithm based on improved k -means and a side window filter (SWF) is designed to achieve an accurate number of shrimp in the lamp house. Specifically, the SWF technique is introduced to preserve the body contour of shrimp larvae, and eliminate noise, such as water impurities and eyes of shrimp larvae. Finally, shrimp larvae are divided into two groups, independent and interdependent, and counted separately. Experimental results show that the designed optical counting system is excellent in terms of visual effect and objective evaluation. © 2023 Optica Publishing Group

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1. INTRODUCTION

Shrimp is an important economic species widely used in aquaculture and fishing. The counting of shrimp larvae is of great significance for maintaining the water quality of aquaculture, monitoring the reproduction of shrimp, and evaluating the effect of scientific research. However, traditional counting methods include visual counting [1], net counting [2], and microscope counting [3], which have a certain simplicity in small-scale farming or research projects. However, with the continued expansion of the shrimp farming scale and technological advances, these methods may encounter some limitations related to the problem of high density [4]. Obviously, the phenomenon of overlap and aggregation of shrimp larvae makes traditional methods by density estimation ineffective. Therefore, it is necessary to explore effective counting methods to improve the accuracy and efficiency of counting.

Automatic counting methods, such as image processing [5], computer vision [6], and flow counting [7], have shown potential to solve the problem of high-density shrimp counting [8]. Furthermore, the light environment plays a crucial role when capturing images inside a container. Factors such as the brightness and uniformity of the light source directly impact the image outcome. For point-light sources and line-light sources [9,10], bright spots negatively affect the quality of images. Therefore, it is necessary to transform point or line light sources into a uniformly illuminated area light source [11] using a backlight configuration, which can be categorized into side-backlight [12]

and direct backlight [13]. The direct backlight offers excellent brightness, but has a relatively large thickness. The latter backlight is compact and energy efficient. To minimize the size of the device, the light emitting diode (LED) is used as point light sources, which are converted to a surface light source with a light guide plate [14] (the structural diagram is shown in Fig. 1).

Recently, there are three types of shrimp larvae counting algorithms, including density estimation [15–18], deep learning based algorithms [8,19–22], and image segmentation [23–25]. The first density estimation [15] algorithm labels the image to generate a corresponding density map [16,17], and then shrimp larvae are counted by estimating the density with various techniques [18]. However, the process of manually labeling each shrimp larva is time consuming. Furthermore, it does not perform well in identifying shrimp larvae. The second algorithm based on deep learning [8,19–22] can obtain accurate identification results with training. Zhang *et al.* designed an automatic shrimp counting method with a lightweight YOLO model [8]. However, this method requires a large number of labeled images, and this algorithm is sensitive to the situation of a small size and a dense distribution of shrimp larvae without training. Wang *et al.* [26] introduced a method to assess the freshness of chilled red shrimp using deep learning. However, the size of the larvae processed by these methods is much larger than that used in our experiments. Additionally, deep learning approaches require the annotation of a large number of images of shrimp larvae to create data sets for training, which requires a significant



Fig. 1. Structural diagram of area light source.

amount of work. Jiang *et al.* [27] introduced an improved point cloud segmentation technique for automated segmentation of mantis shrimp organs. This segmentation is based on spatial, geometric, and textural characteristics to partition the point cloud so that points within the same segment exhibit similar features. However, the shrimp larvae used in our study are quite small with limited distinguishing features, making differentiation challenging. The image segmentation algorithm involves extracting the feature of the targets with morphological operations, and counting shrimp larvae by determining the number of connected regions [23,24]. Solahudin *et al.* utilize the morphological model to separate shrimp from the background with a fixed threshold, and count the shrimp using enhanced images [25]. However, it is not sufficient to divide the image into two categories. Thus, clustering [28] is utilized to effectively separate the shrimp pixel and the background. Furthermore, in the k -means method with a multilevel threshold [29], the classification parameters greatly affect the convergence results [30], and this method cannot determine the specific content and labels of the segmented images. In addition, the accuracy of image segmentation is reduced with water impurities and eyes of shrimp larvae, which needs to be solved.

Based on the above analysis, an automatic counting platform is designed to obtain high-quality images of shrimp larvae. In addition, this paper designs a counting algorithm based on improved k -means. First, a side entry light guide is used as a backlight to ensure a uniform lighting environment for acquiring the shrimp larvae data set. Second, an improved k -means algorithm is designed to extract features and separate the shrimp larvae from background. Furthermore, the side window filter (SWF) model is introduced to suppress the effect of shrimp head and noise [31]. Finally, the shrimp larvae are categorized into two groups, independent and adherent. The experimental result shows that the designed optical system and the counting algorithm can do well in accurately determining the number of shrimp larvae.

The main contributions are described as follows.

- An optical system for shrimp counting is designed using image processing technology.
- The side-type light guide plate is used as the backlight for lighting, reducing the influence of light on imaging.
- A data set referring to shrimp larvae is given to the readers.
- Experimental results prove that the proposed algorithm can identify shrimp larvae with high precision, under the difficult circumstance of partial overlap between large numbers of shrimps.
- The ablation study is given to prove the effectiveness of the side window filter in the proposed method.

In this paper, Section 1 provides a comprehensive overview of current advances in shrimp counting. The proposed framework

is elaborated in Sections 2 and 3. Section 4 gives the experimental analysis. Finally, Section 5 concludes this article.

2. OPTICAL ACQUISITION PLATFORM

The image acquisition system consists of three components, including the camera lenses, the backlight configuration (Section A), and the shrimp container (Section B), which are shown in Figs. 2 and 3.

To ensure the quality of shrimp images, a monochrome industrial camera named Hikvision MV-CE060-10UM with the Hikvision MVL-HF0824M-10MP 1/1.8-inch lens is utilized for real-time image acquisition through the USB3.0 connector. Other parts are described below.

A. Backlight Configuration

The backlight group primarily utilizes a side entry light guide plate, which minimizes the number of required LED lights. As a result, the power consumption and heat generation from the LED light bars are reduced. The light emitted by the LED is

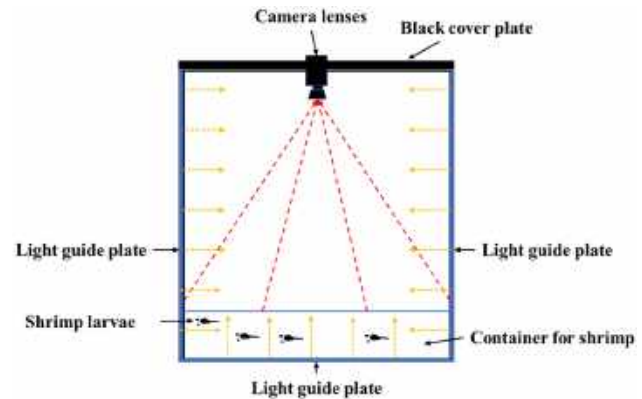


Fig. 2. Designed container with a specific camera.

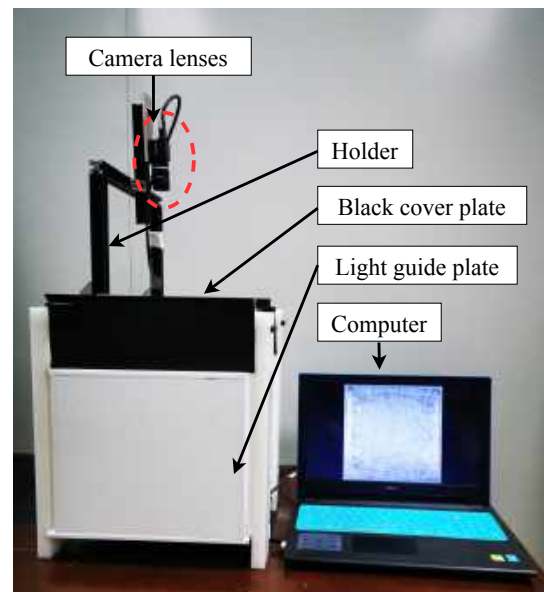


Fig. 3. Optical acquisition platform.

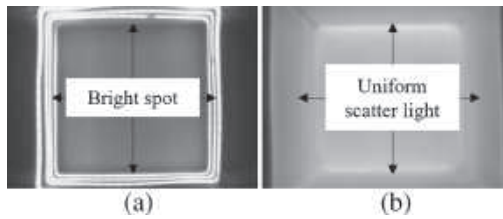


Fig. 4. Comparison results of different light sources. (a) Line light source. (b) Area light source.

coupled into the interior of the light guide plate through its side, following Snell's law:

$$n_1 \sin \theta_1 = n_2 \sin \theta_2, \quad (1)$$

where n_1 and n_2 represent refractive indices of two medias, θ_1 indicates the angle between the incident light and the normal of the interface, and θ_2 is the angle between the refracted light and the normal of the interface.

To disrupt the total internal reflection and achieve a more uniform distribution of light, a regular scattering network is laser engraved beneath the light guide plate. This scattering network ensures that light is directed to specific regions of the plate, resulting in a homogeneous light output. Furthermore, to evaluate the effectiveness of the area light source, both the line light source and the area light source are used as background illumination, to obtain the interior information of the shrimp container, which is shown in Fig. 4. It is evident that Fig. 4(a) exhibits noticeable bright spots, whereas Fig. 4(b) shows a uniform environment. Thus, the area light source helps to create a more uniform lighting environment.

B. Container for Shrimp Larvae

The container of shrimp larvae is an open square bucket with dimensions of 300 mm × 300 mm × 350 mm, which can be seen in Fig. 3. The square bucket is sealed with a lid that has a small hole to install the camera. In addition, the square bucket is made of opalescent polyethylene, which is opaque with the property of good light transmission. This material helps mitigate the impact of ambient light to some extent. In addition, light guide plates are positioned to ensure a uniform lighting environment within the container.

3. PROPOSED AUTOMATIC SHRIMP COUNTING ALGORITHM

A. Work Motivation

Image processing technology finds extensive applications in various domains, offering effective solutions to numerous challenging problems [32]. One such problem is the counting of shrimp fry, which involves determining the number of shrimp larvae present in aquatic environments [6]. As shown in Fig. 5, the structure of the shrimp larvae reveals significant variations in size. With small and highly translucent bodies, shrimp larvae mainly consist of eyes and a head. Due to the high flexibility, shrimp larvae tend to aggregate in local clusters [24], as illustrated in Fig. 6. This phenomenon further exacerbates the difficulty of accurately counting shrimp. In addition, the

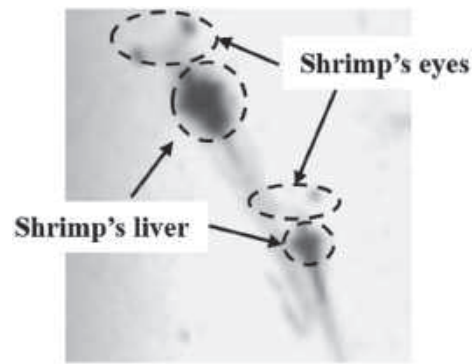


Fig. 5. Structure of shrimp larvae.

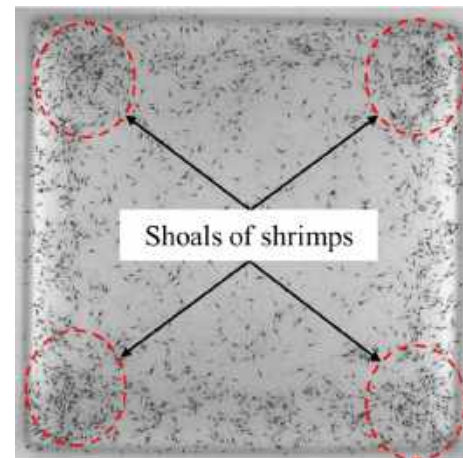


Fig. 6. Group of shrimp larvae.

manual counting method is time consuming and its accuracy is subpar. Recognizing the limitations of traditional approaches, image processing technology has been used in this article, enhancing the accuracy of shrimp larvae counting.

In the experiment, we used 15-day-old South American white shrimp larvae, with heads of 1–2 mm in size. To assess the situation with varying larval densities, we began by photographing around 150 larvae and then incrementally increased their number, with the maximum count reaching about 2000.

Shrimp larvae counting through image segmentation consists of three main stages. First, a pre-processing operation is performed to eliminate noise from the image, and generate a binary representation. Subsequently, morphological operations are applied to the binary image to segment the shrimp larvae. Finally, segmented regions are analyzed using connected domain analysis to obtain final count results [33]. This approach achieves accurate results when shrimp larvae are sparsely distributed and have similar sizes. However, the high density of shrimp larvae makes the accuracy of the counting methods decrease. Specifically, Fig. 7 shows a shrimp image and its corresponding binary image. The experimental result shows that it is difficult to discern the outline of numerous larvae. In addition, adjusting the threshold could result in the exclusion of smaller larvae, which could affect the accuracy of the count of shrimp larvae.

The images of shrimp larvae used in this experiment are captured using the platform designed in this article. With

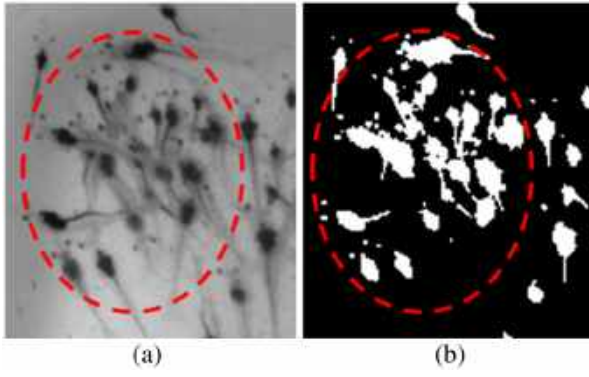


Fig. 7. Image binaryzation. (a) Original image. (b) Binary image.

high-quality images, an algorithm for automatic counting of shrimp larvae is designed to address the limitations of traditional approaches. In a controlled laboratory environment, images of shrimp larvae with varying densities facilitate the evaluation of the performance of the proposed algorithm.

B. Image Pre-processing

Some shrimp larvae at the bottom of the collection box represent weakness in image pixel intensity. Therefore, these larvae can be easily distinguished from the background by enhancing the image contrast. Based on this, subsequent methods can be more effective. Pre-processing of shrimp images is accomplished by combining dynamic stretching and gamma transform, which can enhance both dark and bright details in images [34]. The dynamic stretching method is a linear transformation method that proportionally stretches the image value to enhance its global contrast. In addition, the non-linear gamma transformation model aims to convert the linear output to a non-linear output that matches the perceived luminance of the human eye. Therefore, it enhances the contrast and visual quality of the Image. Given a shrimp image V_{in} , the gamma transformation can be expressed by the following equation:

$$V_{out} = A \cdot (V_{in})^\gamma, \quad (2)$$

where V_{out} denotes the output, γ is gamma parameter, and A is the scaling factor. $\gamma < 1$ refers to the contrast enhancement of dark areas, and $\gamma > 1$ refers to that of bright areas.

To illustrate the impact of the pre-processing technique, a comparison is presented in Fig. 8.

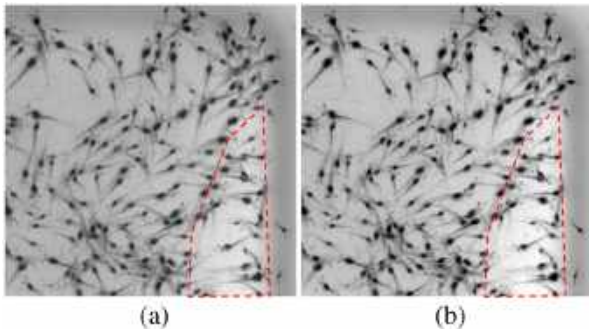


Fig. 8. Contrast enhancement. (a) Original image. (b) Enhanced image.

As shown in Fig. 8, the region enclosed by the red dotted line exhibits a discernibly brighter appearance, while the darker area, such as the head of the shrimp larvae, appears even darker.

C. Image Segmentation

1. Improved K -means Clustering Algorithm

Shrimp larvae of different sizes exhibit variations in pixel values, and the same situation occurs at different water depths. To overcome the limitations associated with threshold selection, the k -means model is employed to cluster the data into k distinct groups, dealing with the issue of incomplete segmentation. Yet, the number of the cluster needs to be set previously. Therefore, an improved k -means algorithm is proposed utilizing a mask model.

Given an image input $D = x_1, x_2, \dots, x_m$, the cluster's number k , and the iteration's maximum number N , the cluster result $C = \{C_1, C_2, \dots, C_k\}$ can be obtained as follows.

- (1) Initialize the cluster of D ; randomly select k samples as central vectors $\{\mu_1, \mu_2, \dots, \mu_k\}$.
- (2) The Euclidean distance between x_i and each central vector $\mu_j (j = 1, 2, \dots, k)$ is calculated by

$$d_{ij} = \|x_i - \mu_j\|^2, \quad (3)$$

where the smallest value of d_{ij} determines the category λ_i to which x_i belongs. Update the cluster assignment as

$$C_{\lambda_i} = C_{\lambda_i} \cup \{x_i\}, \quad (4)$$

where $\cup$ denotes the union symbol.

- (3) Recalculate the central vector μ_j for each cluster C_j :

$$\mu_j = \frac{1}{|C_j|} \sum_{x \in C_j} x. \quad (5)$$

- (4) When all central vectors remain unchanged or the maximum number of iterations N is reached, output the cluster result $C = \{C_1, C_2, \dots, C_k\}$ and go to step (2).
- (5) Evaluate the size of the connected domain in $C = \{C_1, C_2, \dots, C_k\}$, and the cluster C_i with the smallest connected domain is defined as a mask (in Fig. 9), which is utilized to execute the “and” operation with the original shrimp map.

With the mask operation, the classification of shrimp larvae can be effectively determined. Then, the clustering result is shown in Fig. 10.

As shown in Fig. 10(f), the shrimp image is divided into six categories using the masked k -means model. In this paper, we discuss the specific numerical value of k , which is set as 4, 5,



Fig. 9. Effect of the mask operation.

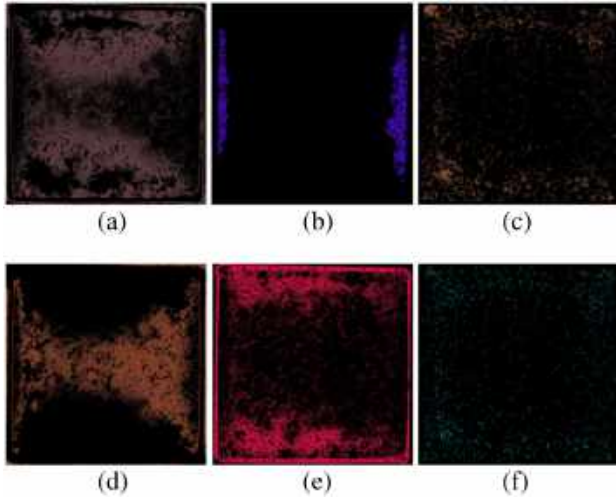


Fig. 10. Clustering result ($k=6$). (a)–(e) Five clusters of background information. (f) Cluster of shrimp larvae.

6, 7, and 8. The experimental results indicated that optimal performance was achieved when k was set to 6, which allowed complete separation of the shrimp larvae from the background. Furthermore, the connected region referring to the shrimp larvae is smaller than that of the background. In short, the masked k -means model helps to accurately obtain the information of shrimp larvae, which is shown in Fig. 9.

2. Side Window Filtering

The eyes and body of shrimp larvae are similar in gray, which affects the masked k -means algorithm in the counting of shrimp larvae. The traditional morphology model employs open operations to remove small noise points, but this model cannot distinguish shrimp with small sizes from the edges of the image. Therefore, the side window filter [35] is utilized to handle the above issue. Furthermore, the masked k -means model can accurately differentiate between the eyes and body of shrimp larvae with the SWF.

Specifically, the SWF enumerates eight possible directions (up, down, left, right, northeast, southeast, southwest, and northwest) for the filtering window. The first four directions place the pixel of interest on one edge of the filtering window, while the latter four directions place the pixel of interest on one corner of the filtering window. Assuming a given pixel, the filtering result $I_i^{r,\rho,\theta}$ can be represented as

$$I_i^{r,\rho,\theta} = F(q_i, r, \rho, \theta), \quad (6)$$

where F denotes the SWF function, q_i represents the input image of the pixel within the filtering window, and r is the radius of the filtering window. $\rho = 0$, r indicates that the pixel is located on the edge and corner of the window, respectively. In addition, $\theta(\pi/2, \pi, 3\pi/2)$ represents the direction of the filtering window.

To preserve edges information, it is necessary to minimize the distance between the input and the output. Therefore, the output I_{SWF} is chosen as the final filtering result, which has the minimum Euclidean distance from the input q_i :

$$I_{SWF} = \arg \min_{\forall I_i^{r,\rho,\theta}} \|q_i - I_i^{r,\rho,\theta}\|_2^2. \quad (7)$$

The output of the current filtering is utilized as the input for the next filtering iteration, and this process is repeated iteratively to achieve enhancement effects. The edge-preserving capability of the SWF model is represented in the ablation section.

D. Overlap Processing and Statistics Counting

With the SWF and the k -means model, most of the segmented shrimp larvae appear as distinct and separate entities. However, some shrimp larvae overlap or stick together in the binary image. In general, the number of shrimp larvae is large and the space in the collection box is small. Thus, the overlap phenomenon will occur as shrimp larvae swim, which is a major consideration for counting shrimp larvae. The result is a fusion of pixels from several shrimp larvae in the captured image, which means the shrimp are sticky. For independent shrimp larvae, the counting task can be performed directly by determining the number of connected domains, which has been validated in previous studies by Khantuwan and Khiripet [36]. Yet, the same counting method can fail when faced with the situation where some shrimp larvae overlap or stick together in the binary image. Therefore, it is necessary to classify shrimp larvae into two types, independent shrimp larvae and sticky shrimp larvae, and obtain the count result separately, which is shown in Fig. 11.

Subsequently, to segment the sticky region between shrimp larvae, an algorithm is designed based on distance transformation [37] and watershed segmentation [38]. Specifically, the distance transformation is utilized to separate sticky objects, obtaining the location of edges. In addition, the watershed algorithm helps to extract the contour, effectively achieving the desired segmentation of sticky shrimp larvae.

According to the above analysis, the proposed counting algorithm consists of five distinct components: preprocessing, enhanced clustering of k -means to extract the contour of shrimp larvae, edge preservation and image denoising, distance transformation and watershed algorithm, and statistical counting. The algorithm flow chart is shown in Fig. 12.

4. EXPERIMENTAL RESULTS AND ANALYSIS

The experimental setup is given in Section A, while Section B presents a qualitative comparison between the proposed algorithm and traditional image segmentation algorithms. Also, a quantitative comparison is provided in Section C. Furthermore, the ablation experiment aims to demonstrate the significance of the SWF model, as discussed in Section D.

A. Experiment Setup

The data set comes from the optical acquisition platform in a controlled laboratory environment. There are 200 images that refer to various densities of approximately 15-day-old South American white shrimp larvae, with a head size ranging from 1 to 2 mm. In fact, the number of pixels for these shrimps ranges from 10 to 130 in images (data and code can be requested by email: kqian@jiangnan.edu.cn). To avoid mixing too much

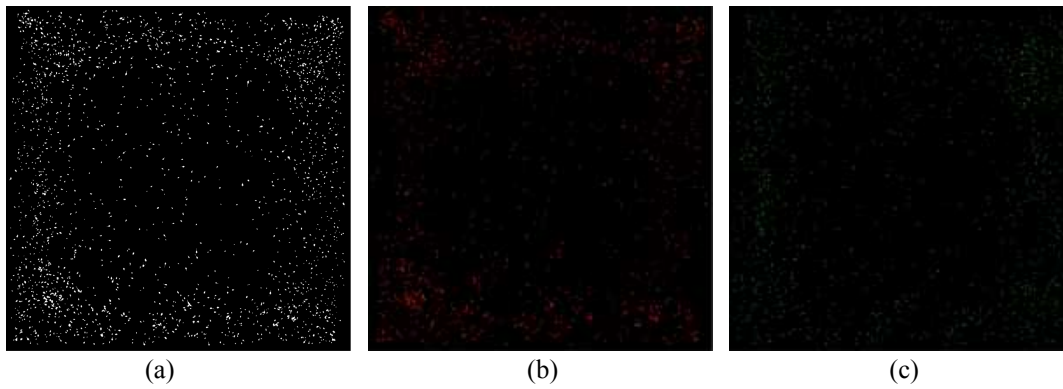


Fig. 11. Classification of shrimp larvae. (a) Segmented image of shrimp larvae. (b) Sticky shrimp larvae. (c) Independent shrimp larvae.

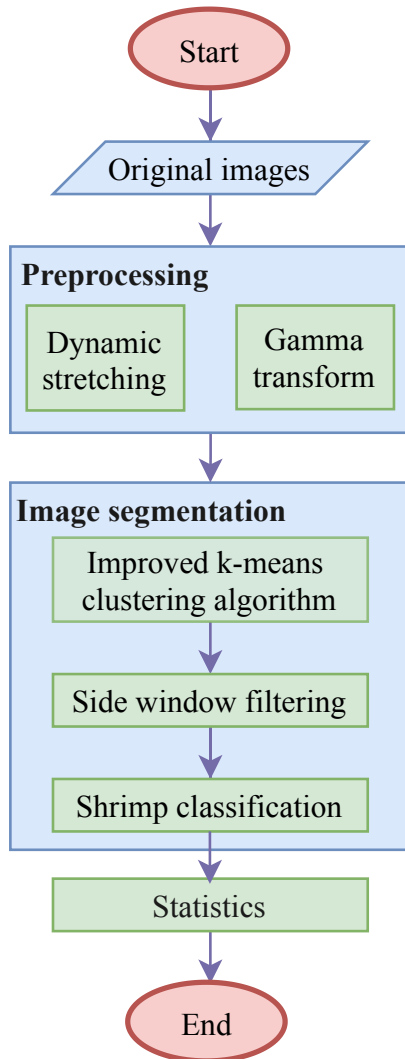


Fig. 12. Flow chart of the proposed algorithm.

background information, areas of interest are cropped to count shrimp larvae. In the k -means model, k is set to 6 for good segmentation performance. Furthermore, the comparisons involve the blob detection algorithm [39], the canny detection algorithm [40], the morphology algorithm (MA) [15], the

k -means, and the proposed masked k -means algorithm. These algorithms are evaluated with different densities of shrimp larvae. The simulation platform is MATLAB-R2020a, and the computer hardware is configured with a 3.2 GHz CPU and 64 Gb of memory. When this evaluation is conducted, a complete understanding of the strengths and weaknesses of each algorithm in the recognition and counting of shrimp larvae can be achieved.

B. Qualitative Comparison

The performance of all algorithms is tested under different densities. The comparison results are presented in Figs. 13–15.

The traditional k -means clustering algorithm successfully identifies shrimp larvae, but fails to filter out noise, such as the eyes of shrimp larvae. The blob detection algorithm, the canny detection algorithm, and the MA are capable of detecting shrimp larvae with large sizes, yet the accuracy of these algorithms is low referring to small sizes. Moreover, these algorithms do not perform well in segmenting shrimp larvae that are sticky and overlapped with each other, which is presented in Fig. 15.

In general, the proposed algorithm demonstrated superior performance across various densities, as shown in Figs. 13–15. Even facing small-sized shrimp, the proposed algorithm performs well in obtaining the accurate number of shrimp larvae.

C. Quantitative Comparison

For further evaluation of the performance of comparison algorithms, the error rate (ER) is utilized as a measure. Specifically, the count accuracy is calculated by comparing the estimated number I_d with the actual number A_c referring to shrimp larvae. And the error rate E is obtained by one minus the counting accuracy:

$$E = |I_d - A_c| / A_c. \quad (8)$$

Subsequently, the ER of the blob, the canny, the MA, the k -means, and the proposed masked k -means referring to various densities is shown in Table 1. The numbers in parentheses following the “data” in the table represent the actual count of shrimp larvae.

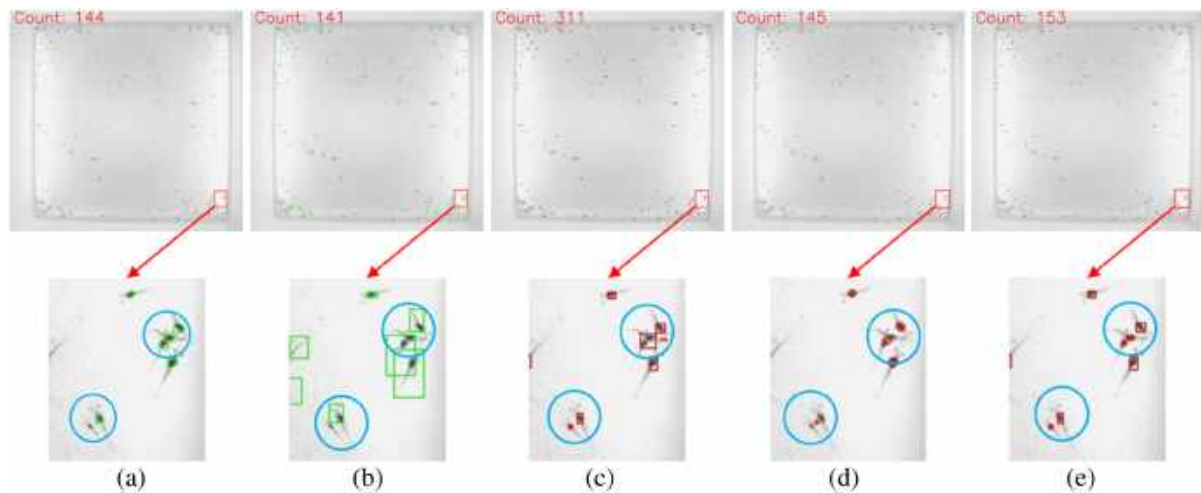


Fig. 13. Comparison results in the case of sparse density (151). (a) MA (144). (b) Canny (141). (c) K -means (311). (d) Blob (145). (e) Masked k -means (153).

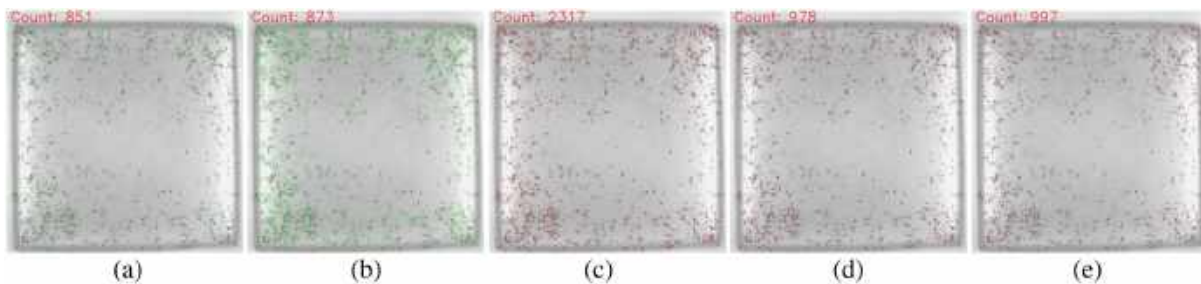


Fig. 14. Comparison results in the case of medium density (1005). (a) MA (851). (b) Canny (873). (c) K -means (2317). (d) Blob (978). (e) Masked k -means (997).

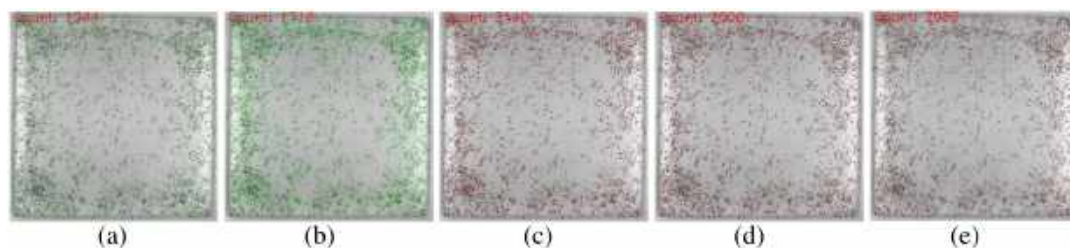


Fig. 15. Comparison results in the case of dense density (2135). (a) MA (1244). (b) Canny (1718). (c) K -means (3140). (d) Blob (2008). (e) Masked k -means (2089).

Table 1. Comparison of the ER

Data	Blob	Canny	MA	K -means	Masked K -means
Data1 (151)	3.97%	6.62%	4.63%	105.96%	1.32%
Data2 (449)	3.56%	10.02%	8.46%	99.35%	0.89%
Data3 (1005)	2.68%	13.13%	15.32%	130.54%	0.79%
Data4 (2135)	5.94%	19.53%	41.73%	47.07%	2.15%

In Table 1, the algorithm proposed in this manuscript has an error rate of around 1% when the number of shrimp larvae is small and the density is low. As the density of larvae increases and the number reaches approximately 1000, the count error rate remains at approximately 1%, ensuring accurate counting. When the density continues to rise, reaching around 2000

larvae, the counting error is about 2%. Although the error rate has increased by one, it still allows accurate counting. It indicates that both the proposed algorithm and other algorithms perform well in counting shrimp larvae with a small number. However, the ER E of the proposed algorithm is consistently lower than that of other algorithms. As the number of shrimp larvae increases, the count accuracy decreases accordingly. It should be mentioned that the ER of the proposed algorithm remains below 5%, which is superior to other algorithms.

In conclusion, the combination of subjective evaluation and objective evaluation confirms the effectiveness of the proposed algorithm in achieving accurate recognition and counting of shrimp larvae.

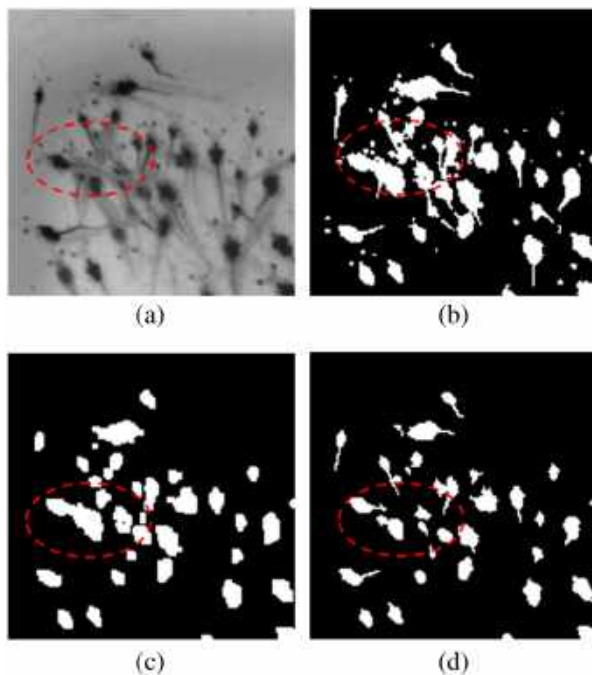


Fig. 16. Ablation results in the case of dense shrimp. (a) Original image. (b) Result without filtering. (c) Result with the opening operation. (d) Processing result with SWF.

D. Ablation Study

To verify the effectiveness of the SWF model, the ablation experiment is conducted, which is shown in Figs. 16 and 17. Initially, the masked k -means without any filtering processing is selected as the baseline. Then, two additional variants are developed by incorporating the side window filtering or morphological operations. These three algorithms are applied individually to process images that contain dense or sparse shrimp larvae.

As can be seen in Figs. 16 and 17, it is evident that the basic algorithm without filtering processing introduces a significant amount of noise, while the inclusion of filtering processing improves the overall performance. Additionally, the SWF model outperforms open operation in terms of edge preservation and noise suppression. In particular, open operation tends to filter out small shrimp larvae as noise. In fact, the SWF achieves good results by effectively retaining the contour of the shrimp larvae while removing noise.

E. Discussion

With the shrimp larvae image capture platform, images of larvae can be obtained at various densities and quantities. In addition, the number of larvae present can be calculated with image processing technology. As a result, the experimental results highlight the superior capabilities of the side window filtering model, especially in scenarios involving shrimp larvae with high density. To be specific, the proposed algorithm demonstrates excellent counting performance when the number of shrimp larvae is less than 2000, with an accuracy rate of over 0.97, which outperforms other similar counting algorithms.

However, the parameter of the k -means method needs to be set in advance, which prompts us to choose non-parametric

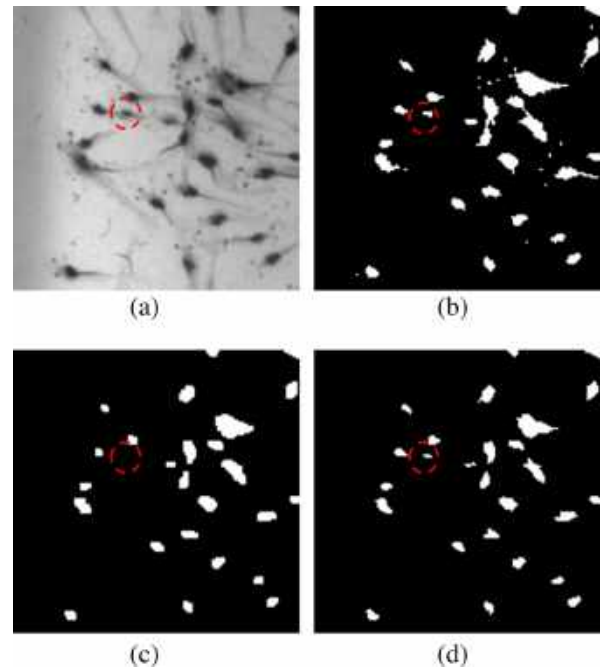


Fig. 17. Ablation results in the case of sparse shrimp. (a) Original image. (b) Result without filtering. (c) Result with the opening operation. (d) Processing result with SWF.

clustering methods in future work. In addition, better annotated databases are obtained with this platform, and the deep learning technology can be utilized to extract the features of small shrimp.

5. CONCLUSIONS

In this paper, an optical platform is designed and a corresponding algorithm is proposed for counting shrimp larvae. The image acquisition platform helps to obtain high-quality images, which will be public. In addition, the side window filter approach works well in preserving the contour of shrimp larvae and suppressing the influence of the shrimp eye. Moreover, overlap processing effectively addresses the situation of sticky shrimp larvae. This sets the foundation for developing future small target counting in aquaculture and fishing.

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Data availability. Data underlying the results presented in this paper are not publicly available at this time but may be obtained from the authors upon reasonable request.

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